

.	.	<i>B</i>	.	.
.	<i>F</i>	.	.	<i>A</i>
.	<i>C</i>	.	.	.
<i>D</i>	.	.	<i>G</i>	.
<i>E</i>	.	.	.	<i>X</i>

In this example, the closest letter is G, and the farthest is B.

3.4 Word Search

In this task, LLMs were provided a programmatically generated grid containing a single word along with the word contained. Models were tasked with producing a list of coordinates identifying the word’s location. Credit was given for partial spellings and incorrect orderings of coordinates. An example is shown below:

<i>D</i>	<i>A</i>	<i>E</i>	<i>Z</i>	<i>A</i>
<i>E</i>	<i>D</i>	<i>B</i>	<i>V</i>	<i>D</i>
<i>R</i>	<i>A</i>	<i>T</i>	<i>M</i>	<i>M</i>
<i>L</i>	<i>T</i>	<i>C</i>	<i>Z</i>	<i>Y</i>
<i>J</i>	<i>J</i>	<i>N</i>	<i>N</i>	<i>L</i>

In this example, the word “RAT” is embedded horizontally in the third row.

3.5 Slide

In this task, models were given a grid with an X and numerous #s. The model was instructed to ‘slide’ the X until it hit either a wall (#) or the edge of the grid and output the final coordinates. For multi-slide problems, credit was given for correct secondary slides, even if the starting position was affected from an earlier slide. An example grid is show below:

.	#	#	.	.	.	#	#	.	.	.	#	#	.	.
.	#	.	<i>X</i>	.	.	#	.	.	.	.	#	.	.	.
.	.	.	.	.	→	.	.	.	<i>X</i>	→	<i>X</i>	.	.	.
.	#	#	#	.	.	#	#	#	.	.	#	#	#	.
#	.	.	.	#	#	.	.	.	#	#	.	.	.	#

Figure 4: Slide down, then left

In this example, the X would first move to row 2, column 3 and then end at row 2, column 0.

4 Results

Each of the designed tasks was tested on four models: GPT-4o [9], GPT-4.1 [10], Claude 3.7 No Thinking, and Claude 3.7 Medium Thinking [2]. Each set of parameters (i.e. grid size, model) was run 10 times, and the results were averaged. Test prompt inputs ranged from ~400 to ~90,000 tokens, well below the context limit for any of the models.

Additionally, we explored some common sources of error and attempted to mitigate them. These attempts are also seen below.

4.1 Quadrant

Models	Smallest 5 Grids	Largest 5 Grids
GPT-4o	80%	22%
GPT-4.1	86%	35%
Claude No Thinking	98%	61%
Claude Medium Thinking	98%	58%

Table 1: Model Accuracy on Small and Large Grids (Quadrant)

Grid sizes ranged from 2×2 to 300×300 in increments of 10×10 . Small grids saw high accuracy across all models, but performance deteriorated significantly on larger grids. Specifically, the OpenAI variants performed poorly on larger grids, with accuracy dropping to 20-30%, and Anthropic models dropping to around 60%, as seen in Figure 6.

Error analysis revealed that while initial misreading of the grid was common, models were largely consistent in quadrant selection based on their internal understanding of the board state. That is to say, once the model had “found” the X in the grid, it predicted the correct quadrant for that coordinate position; errors appeared largely when the model identified the incorrect position as containing the X. This trend is shown in Figure 7.

The models were internally inconsistent only when the X was near a dividing line between two quadrants. Even when given the board size explicitly, models often had a difficult time establishing the location of the center lines. The increase in consistency at larger board sizes can likely be explained by the lower chance of an X placement near one of the dividing lines.

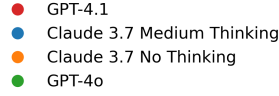


Figure 5: Color coding of data; all following plots use this format

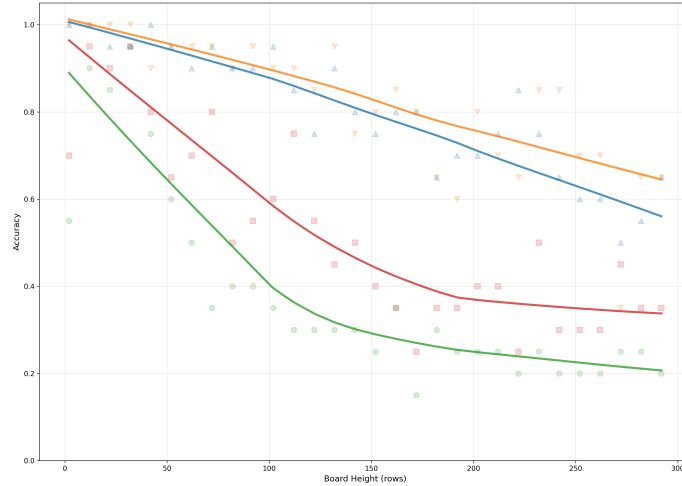


Figure 6: Accuracy vs Grid Size (Quadrant)

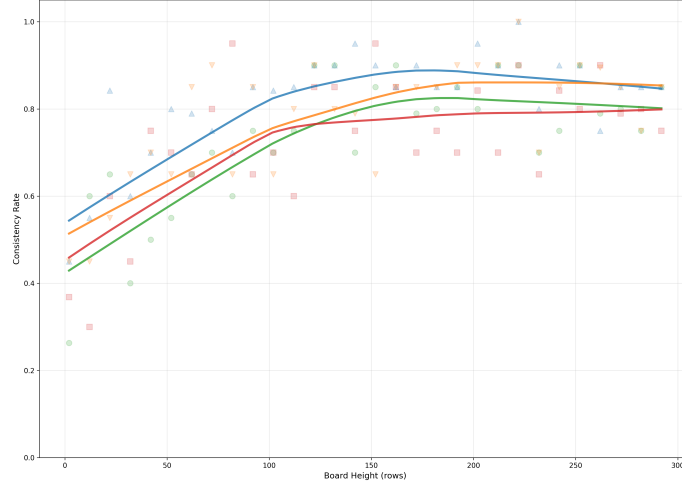


Figure 7: Consistency vs Grid Size (Quadrant)

4.2 Transformation

Models	Smallest 5 Grids	Largest 5 Grids
GPT-4o	6%	0%
GPT-4.1	14%	0%
Claude No Thinking	62%	0%
Claude Medium Thinking	70%	0%

Table 2: Model Accuracy on Small and Large Grids (Transform)

Our suspicions about errors from the Quadrant test stemming from the models misreading the initial board were quickly proven in the Transformation test. Grid sizes ranged from 2×2 to 200×200 , with each grid testing reflections across both axes. The models’ inability to properly read board states was immediately evident: even on small boards, the OpenAI models averaged under 50% accuracy. Performance deteriorated much quicker as well, with Anthropic models dropping below 20% accuracy by the 75×75 grid and the OpenAI models degrading even faster. By the largest grids, models rarely, if ever, got the answer correct. These trends can be seen in Figure 8.

It seems that the margin of error built into the nature of the Quadrant test shielded the models from such immediate deterioration. Interestingly, the models were not as internally consistent in this test as well. Both Anthropic models performed very well, with perfect internal consistency whereas the OpenAI models sometimes hit as low as 20% calculation accuracy for certain grids. There appears to be no definite pattern for which grids were inconsistent, and the results can be seen in Figure 9.